

The Alternate Prescription

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Abstract

1.

$$\begin{aligned}
 & : \Pi^m A_m^i : (z_i) : \Pi^m A_m^j : (z_j) \\
 & = - \frac{(k_j \cdot A_i) (k_i \cdot A_j) + (A_i \cdot A_j)}{z_{ij}^2} - \frac{(k_j \cdot \partial A_i) (k_i \cdot A_j) + (\partial A_i \cdot A_j)}{z_{ij}} \\
 & \quad + \frac{: \Pi^m A_m^i (k_i \cdot A_j) : - : \Pi^m (k_j \cdot A_i) A_m^j :}{z_{ij}}
 \end{aligned} \tag{1} \{?\}$$

2.

$$\begin{aligned}
 & : \Pi^m A_m^i : (z_i) : d_\alpha W_j^\alpha : (z_j) \\
 & = \frac{D_\alpha (k_j \cdot A^i) W_j^\alpha}{z_{ij}^2} + \frac{\partial (D_\alpha (k_j \cdot A^i)) W_j^\alpha}{z_{ij}} \\
 & \quad - \frac{: \Pi^m D_\alpha A_m^i W_j^\alpha : + : d_\alpha (k_j \cdot A_i) W_j^\alpha : + : \partial \theta^\alpha A_m^i (\gamma^m W_j)_\alpha :}{z_{ij}}
 \end{aligned} \tag{2} \{?\}$$

3.

$$: \Pi^m A_m^i : (z_i) : \partial \theta^\alpha A_\alpha^2 : (z_j) = - \frac{\partial \theta^\alpha (k_j \cdot A_i) A_\alpha^j}{z_{ij}} \tag{3} \{?\}$$

4.

$$: \Pi^m A_m^i : (z_i) \frac{1}{2} : N^{pq} F_{pq}^j : (z_j) = - \frac{1}{2} \frac{: N^{mn} (k_j \cdot A_i) F_{mn}^j :}{z_{ij}} \tag{4} \{?\}$$

5.

$$\begin{aligned}
& : d_\alpha W_i^\alpha : (z_i) : \Pi^m A_m^j : (z_j) \\
& = - \frac{W_i^\alpha D_\alpha (k_i \cdot A_j)}{z_{ij}^2} - \frac{\partial W_i^\alpha D_\alpha (k_i \cdot A_j)}{z_{ij}} \\
& - \frac{: \Pi^m W_i^\alpha D_\alpha A_m^j : - : d_\alpha W_i^\alpha (k_i \cdot A_j) : - : \partial \theta^\alpha (\gamma^m W_i)_\alpha A_m^j :}{z_{ij}}
\end{aligned} \tag{5} \{?\}$$

6.

$$\begin{aligned}
& : d_\alpha W_i^\alpha : (z_i) : d_\beta W_j^\beta : (z_j) \\
& = \frac{D_\beta W_i^\alpha D_\alpha W_j^\beta + k_m^{ij} (W_i \gamma^m W_j)}{z_{ij}^2} + \frac{\partial (D_\beta W_i^\alpha) D_\alpha W_j^\beta + k_m^{ij} (\partial W_i \gamma^m W_j)}{z_{ij}} \\
& + \frac{: \Pi^m (W_i \gamma_m W_j) : + : d_\alpha (D_\beta W_i^\alpha W_j^\beta - W_i^\beta D_\beta W_j^\alpha) :}{z_{ij}}
\end{aligned} \tag{6} \{?\}$$

7.

$$: d_\alpha W_i^\alpha : (z_i) : \partial \theta^\beta A_\beta^j : (z_j) = - \frac{(W_i A_j)}{z_{ij}^2} - \frac{(\partial W_i A_j)}{z_{ij}} - \frac{: \partial \theta^\alpha W_i^\beta D_\beta A_\alpha^j :}{z_{ij}} \tag{7} \{?\}$$

8.

$$: d_\alpha W_i^\alpha : (z_i) \frac{1}{2} : N^{mn} F_{mn}^j : (z_j) = - \frac{1}{2} \frac{N^{mn} W_i^\alpha D_\alpha F_{mn}^j}{z_{ij}} \tag{8} \{?\}$$

9.

$$: \partial \theta^\alpha A_\alpha^i : (z_i) : \Pi^m A_m^j : (z_j) = \frac{: \partial \theta^\alpha A_\alpha^i (k_i \cdot A_j) :}{z_{ij}} \tag{9} \{?\}$$

10.

$$: \partial \theta^\alpha A_\alpha^i : (z_i) : d_\alpha W_j^\alpha : (z_j) = \frac{(A^i W_j)}{z_{ij}^2} + \frac{(\partial A^i W_j)}{z_{ij}} + \frac{: \partial \theta^\alpha D_\beta A_\alpha^i W_j^\beta :}{z_{ij}} \tag{10} \{?\}$$

11.

$$: \partial \theta^\alpha A_\alpha^i : (z_i) : \partial \theta^\beta A_\beta^j : (z_j) = 0 \tag{11} \{?\}$$

12.

$$: \partial \theta^\alpha A_\alpha^i : (z_i) \frac{1}{2} : N^{mn} F_{mn}^j : (z_j) = 0 \quad (12) \{?\}$$

13.

$$\frac{1}{2} : N^{mn} F_{mn}^i : (z_i) : \Pi^p A_p^j : (z_j) = \frac{1}{2} \frac{N^{mn} F_{mn}^i (k_i \cdot A_j) :}{z_{ij}} \quad (13) \{?\}$$

14.

$$\frac{1}{2} : N^{mn} F_{mn}^i : (z_i) : d_\alpha W_j^\alpha : (z_j) = -\frac{1}{2} \frac{N^{mn} D_\alpha F_{mn}^i W_j^\alpha :}{z_{ij}} \quad (14) \{?\}$$

15.

$$\frac{1}{2} : N^{mn} F_{mn}^i : (z_i) : \partial \theta^\alpha A_\alpha^j : (z_j) = 0 \quad (15) \{?\}$$

16.

$$\frac{1}{2} : N^{mn} F_{mn}^i : (z_i) \frac{1}{2} : N^{pq} F_{pq}^j : (z_j) = \frac{3}{2} \frac{F_{mn}^i F_j^{mn}}{z_{ij}^2} + \frac{3}{2} \frac{\partial F_{mn}^i F_j^{mn}}{z_{ij}} - \frac{N^{mn} F_{mp}^i F_j^p}{z_{ij}} \quad (16) \{?\}$$

We make the observation that the terms in the blue are the worldsheet are same as the terms in the quadratic pole except that the superfield with label i appear with worldsheet derivative. These can be re-written as

$$\partial K(X, \theta) = \Pi^m k_m K(X, \theta) + \partial \theta^\alpha D_\alpha K(X, \theta) \quad (17) \{?\}$$

The OPE can be written as follows

$$U_{ij} \equiv \frac{\tilde{U}_{ij}}{z_{ij}^2} + \frac{1}{z_{ij}} \left(: \Pi^m A_m^{ij} : + : d_\alpha W_{ij}^\alpha : + : \partial \theta^\alpha A_\alpha^{ij} : + \frac{1}{2} : N^{mn} F_{mn}^{ij} : \right) \equiv \frac{\tilde{U}_{ij}}{z_{ij}^2} + \frac{\hat{U}_{ij}}{z_{ij}} \quad (18) \{?\}$$

where,

$$\hat{U}_{ij} \equiv : \Pi^m A_m^{ij} : + : d_\alpha W_{ij}^\alpha : + : \partial \theta^\alpha A_\alpha^{ij} : + \frac{1}{2} : N^{mn} F_{mn}^{ij} : \quad (19) \{?\}$$

is the integrated vertex operator for the multi-particle fields.

$$\begin{aligned}\tilde{U}_{ij} = & -(k_j \cdot A_i)(k_i \cdot A_j) - (A_i \cdot A_j) + D_\alpha(k_j \cdot A^i)W_j^\alpha - W_i^\alpha D_\alpha(k_i \cdot A_j) \\ & + D_\beta W_i^\alpha D_\alpha W_j^\beta + k_m^{ij}(W_i \gamma^m W_j) - (W_i A_j) + (A^i W_j) + \frac{3}{2}F_{mn}^i F_j^{mn}\end{aligned}\quad (20) \{?\}$$

By making use of the equation of motions, we can write this in a very simple form

$$\begin{aligned}\tilde{U}_{ij} = & -(A_i \cdot A_j) - (k_i \cdot k_j)(A_i \cdot A_j) + (k_j \cdot k_i)(A^i W_j) \\ & - (k_i \cdot k_j)(W^i A_j) - (W_i A_j) + (A^i W_j) \\ = & (1 + (k_i \cdot k_j))[(A^i W_j) - (W_i A_j) - (A_i \cdot A_j)]\end{aligned}\quad (21) \{?\}$$

Note that this can be written as

$$\begin{aligned}\tilde{U}_{ij} = & -\frac{1}{2}(k_j \cdot A_i)(k_i \cdot A_j) - \frac{1}{2}(A_i \cdot A_j) + D_\alpha(k_j \cdot A^i)W_j^\alpha + \frac{1}{2}D_\beta W_i^\alpha D_\alpha W_j^\beta \\ & + \frac{1}{2}k_m^{ij}(W_i \gamma^m W_j) - (W_i A_j) + \frac{3}{4}F_{mn}^i F_j^{mn} + (i \leftrightarrow j)\end{aligned}\quad (22) \{?\}$$

where, we show that $k_m^i \tilde{U}^{ij}$ is vanishes when we are considering three point amplitudes as we shall show later.

Next we collect all the structures associated with Π^m and call them A_m^{ij} i.e.

$$A_m^{ij} \equiv A_m^i(k_i \cdot A_j) - (k_j \cdot A_i)A_m^j - D_\alpha A_m^i W_j^\alpha - W_i^\alpha D_\alpha A_m^j + (W_i \gamma_m W_j) + k_m^i \tilde{U}^{ij}\quad (23) \{?\}$$

Similarly,

$$W_j^\alpha \equiv -(k_j \cdot A_i)W_j^\alpha + W_i^\alpha(k_i \cdot A_j) + D_\beta W_i^\alpha W_j^\beta - W_i^\beta D_\beta W_j^\alpha\quad (24) \{?\}$$

$$A_\alpha^{ij} \equiv -A_m^i(\gamma^m W_j)_\alpha - (k_j \cdot A_i)A_\alpha^j + (\gamma^m W_i)_\alpha A_m^j - W_i^\beta D_\beta A_\alpha^j + A_\alpha^i(k_i \cdot A_j) + D_\beta A_\alpha^i W_j^\beta + D_\alpha^i \tilde{U}^{ij}\quad (25) \{?\}$$

$$F_{mn}^{ij} \equiv -(k_j \cdot A_i)F_{mn}^j - W_i^\alpha D_\alpha F_{mn}^j + F_{mn}^i(k_i \cdot A_j) - D_\alpha F_{mn}^i W_j^\alpha - 2F_{mp}^i F_j^p{}_n\quad (26) \{?\}$$

Let us now calculate the OPE of U_1 with that of U_{23} i.e.

$$\overline{U_1 U_{23}} = \frac{\overline{U_1(z_1) \tilde{U}^{23}(z_3)}}{z_{23}^2} + \frac{\overline{U_1(z_1) \hat{U}^{23}(z_3)}}{z_{23}}\quad (27) \{?\}$$

1 Application to three point computation

The amplitude will be given by the terms that are at cubic order in the singularity. Thus, we need to pick the single pole term in the OPE of $U_1 \tilde{U}_{23}$ and the quadratic piece in the OPE of $U_1 \hat{U}_{23}$. So, far we have not made use of the sYM e.o.m. But, now we make use of these and the kinematic constraints for massless states in three point amplitudes i.e. $k_i \cdot k_j = 0$. For the first case we have

$$\begin{aligned}
U_1(z_1) \tilde{U}^{23}(z_3) &= : \Pi^m A_m^1 : \tilde{U}^{23}(z_3) + : d_\alpha W_1^\alpha : \tilde{U}^{23}(z_3) \\
&= \frac{-k_{23}^m A_m^1 \tilde{U}^{23}}{z_{13}} - \frac{W_1^\alpha D_\alpha \tilde{U}^{23}}{z_{13}} + \text{regular} \\
&= \frac{k_1^m A_m^1 \tilde{U}^{23}}{z_{13}} - \frac{W_1^\alpha D_\alpha \tilde{U}^{23}}{z_{13}} + \text{regular}
\end{aligned} \tag{28} \{?\}$$

Hence the relevant piece for the amplitude is

$$S_{1(23)}^I \equiv - \frac{W_1^\alpha D_\alpha \tilde{U}^{23}}{z_{13} z_{23}^2} \tag{29} \{?\}$$

Further using the result in (??) we find that by noting ($i \rightarrow 1, j \rightarrow 23$)

$$\begin{aligned}
\frac{U_1(z_1) \hat{U}^{23}(z_3)}{z_{23}} &= \frac{1}{z_{23} z_{13}^2} \left[-(\mathbf{k}_{23} \cdot \mathbf{A}_1) (k_1 \cdot A_{23}) - (A_1 \cdot A_{23}) + D_\alpha (\mathbf{k}_{23} \cdot \mathbf{A}^1) W_{23}^\alpha - W_1^\alpha D_\alpha (k_1 \cdot A_{23}) \right. \\
&\quad \left. + D_\beta W_1^\alpha D_\alpha W_{23}^\beta + \mathbf{k}_m^{23} (W_1 \gamma^m W_{23}) - (W_1 A_{23}) + (A^1 W_{23}) + \frac{3}{2} F_{mn}^1 F_{23}^{mn} \right] \\
&= \frac{1}{z_{23} z_{13}^2} \left[- (A_1 \cdot A_{23}) - W_1^\alpha D_\alpha (k_1 \cdot A_{23}) + D_\beta W_1^\alpha D_\alpha W_{23}^\beta - (W_1 A_{23}) \right. \\
&\quad \left. + (A^1 W_{23}) + \frac{3}{2} F_{mn}^1 F_{23}^{mn} \right]
\end{aligned} \tag{30} \boxed{\text{contri_}}$$

This, contribution we shall call $S_{1(23)}^{II}$ and thus, we need to calculate

$$\begin{aligned}
S_{1(23)}^{II} &\equiv \frac{1}{z_{23} z_{13}^2} \left[- (A_1 \cdot A_{23}) - W_1^\alpha D_\alpha (k_1 \cdot A_{23}) + D_\beta W_1^\alpha D_\alpha W_{23}^\beta - (W_1 A_{23}) \right. \\
&\quad \left. + (A^1 W_{23}) + \frac{3}{2} F_{mn}^1 F_{23}^{mn} \right]
\end{aligned} \tag{31} \{?\}$$

Let us calculate each of the contributions one by one.

1.1 $S_{1(23)}^I$

We note that

$$\begin{aligned}
D_\alpha \tilde{U}^{ij} &= D_\alpha \left(-(k_j \cdot A_i) (k_i \cdot A_j) - (A_i \cdot A_j) + D_\beta (k_j \cdot A^i) W_j^\beta - W_i^\beta D_\beta (k_i \cdot A_j) \right) \\
&\quad + D_\alpha \left(D_\beta W_i^\sigma D_\sigma W_j^\beta + k_m^{ij} (W_i \gamma^m W_j) - (W_i A_j) + (A^i W_j) + \frac{3}{2} F_{mn}^i F_j^{mn} \right) \\
&= -(k_j \cdot D_\alpha A_i) (k_i \cdot A_j) - (k_j \cdot A_i) (k_i \cdot D_\alpha A_j) - (D_\alpha A_i \cdot A_j) - (A_i \cdot D_\alpha A_j) \\
&\quad + D_\alpha D_\beta (k_j \cdot A^i) W_j^\beta - D_\beta (k_j \cdot A^i) D_\alpha W_j^\beta - D_\alpha W_i^\beta D_\beta (k_i \cdot A_j) + W_i^\beta D_\beta (k_i \cdot D_\alpha A_j) \\
&\quad + D_\alpha D_\beta W_i^\sigma D_\sigma W_j^\beta + D_\beta W_i^\sigma D_\alpha D_\sigma W_j^\beta - (D_\alpha W_i A_j) + (W_i D_\alpha A_j) \\
&\quad + (D_\alpha A^i W_j) - (A^i D_\alpha W_j) + \frac{3}{2} D_\alpha F_{mn}^i F_j^{mn} + \frac{3}{2} F_{mn}^i D_\alpha F_j^{mn} \\
&= -k_m^j (\gamma^m W_i)_\alpha (k_i \cdot A_j) - (k_j \cdot A_i) k_m^i (\gamma^m W_j)_\alpha - (\gamma^m W_i)_\alpha A_m^j - A_\alpha^i (k_i \cdot A_j) \\
&\quad - A_m^i (\gamma^m W_j)_\alpha - (k_j \cdot A_i) A_\alpha^j + \frac{1}{4} k_m^j F_{st}^i (\gamma^{st} \gamma^m W_j)_\alpha - \frac{1}{4} k_m^j (\gamma^{st} \gamma^m W_i)_\alpha F_{st}^j \\
&\quad - \frac{1}{4} F_{st}^i k_m^i (\gamma^{st} \gamma^m W_j)_\alpha - \frac{1}{4} k_m^i (\gamma^m \gamma^{st} W_i)_\alpha F_{st}^j + 4k_j^t (\gamma_t W^i)_\alpha (k_i \cdot A_j) \\
&\quad - \frac{1}{4} F_{mn}^i (\gamma^{mn} A_j)_\alpha + (W_i D_\alpha A_j) + (D_\alpha A^i W_j) - \frac{1}{4} (\gamma_{mn} A_i)_\alpha F_{mn}^i \\
&\quad + 3k_m^i (\gamma_n W_i)_\alpha F_j^{mn} + 3F_{mn}^i k_m^j (\gamma^n W_j)_\alpha
\end{aligned} \tag{32} \{?\}$$

Thus, we find that

$$\begin{aligned}
W^\alpha D_\alpha \tilde{U}^{23} &= -k_m^3 (W_1 \gamma^m W_2) (k_2 \cdot A_3) - (k_3 \cdot A_2) k_m^2 (W_1 \gamma^m W_3) - (W_1 \gamma^m W_2) A_m^3 - (W_1 A^2) (k_2 \cdot A_3) \\
&\quad - A_m^2 (W_1 \gamma^m W_3) - (k_3 \cdot A_2) (W^1 A^3) + \frac{1}{4} k_m^3 F_{st}^2 (W_1 \gamma^{st} \gamma^m W_3) - \frac{1}{4} k_m^3 (W_1 \gamma^{st} \gamma^m W_2) F_{st}^3 \\
&\quad - \frac{1}{4} F_{st}^2 k_m^2 (W_1 \gamma^{st} \gamma^m W_3) - \frac{1}{4} k_m^2 (W_1 \gamma^m \gamma^{st} W_2) F_{st}^3 + 4k_3^t (W_1 \gamma_t W^2) (k_2 \cdot A_3) \\
&\quad - \frac{1}{4} F_{mn}^2 (W_1 \gamma^{mn} A_3) + W_1^\alpha (W_2 D_\alpha A_3) + W_1^\alpha (D_\alpha A^2 W_3) + \frac{1}{4} (W_1 \gamma_{mn} A_2) F_{mn}^3 \\
&\quad + 3k_m^2 (W_1 \gamma_n W_2) F_3^{mn} + 3F_{mn}^2 k_3^m (W_1 \gamma^n W_3) \\
&= -(W_1 \gamma^m W_2) A_m^3 - (W_1 A^2) (k_2 \cdot A_3) - A_m^2 (W_1 \gamma^m W_3) - (k_3 \cdot A_2) (W^1 A^3) \\
&\quad - \frac{1}{4} F_{mn}^2 (W_1 \gamma^{mn} A_3) + W_1^\alpha (W_2 D_\alpha A_3) + W_1^\alpha (D_\alpha A^2 W_3) - \frac{1}{4} (W_1 \gamma_{mn} A_2) F_{mn}^3
\end{aligned} \tag{33} \{?\}$$

where, the last line vanishes due to

$$k_m^2 (W_1 \gamma_n W_2) F_3^{mn} = (k_2 \cdot k_3) (W_1 \gamma_n W_2) A_3^n - k_3^n (W_1 \gamma_n W_2) = 0 \tag{34} \{?\}$$

Thus, we have

$$\begin{aligned}
-W_1^\alpha D_\alpha \tilde{U}^{23} &= (W_1 \gamma^m W_2) A_m^3 + (W_1 A^2) (k_2 \cdot A_3) + A_m^2 (W_1 \gamma^m W_3) + (k_3 \cdot A_2) (W^1 A^3) \\
&\quad + \frac{1}{4} F_{mn}^2 (W_1 \gamma^{mn} A_3) - W_1^\alpha (W_2 D_\alpha A_3) - W_1^\alpha (D_\alpha A^2 W_3) + \frac{1}{4} (W_1 \gamma_{mn} A_2) F_{mn}^3
\end{aligned}$$

Thus, we find that

$$\begin{aligned}
S_{1(23)}^I &= \frac{1}{z_{13} z_{23}^2} \left[(W_1 \gamma^m W_2) A_m^3 + (W_1 A^2) (k_2 \cdot A_3) + A_m^2 (W_1 \gamma^m W_3) + (k_3 \cdot A_2) (W^1 A^3) \right. \\
&\quad \left. + \frac{1}{4} F_{mn}^2 (W_1 \gamma^{mn} A_3) - W_1^\alpha (W_2 D_\alpha A_3) - W_1^\alpha (D_\alpha A^2 W_3) + \frac{1}{4} (W_1 \gamma_{mn} A_2) F_{mn}^3 \right] \quad (35) \{?\}
\end{aligned}$$

1.2 $S_{1(23)}^{II}$

To calculate this we need to know

$$\begin{aligned}
D_\alpha^i \tilde{U}^{ij} &= -(k_j \cdot D_\alpha A_i) (k_i \cdot A_j) - (D_\alpha A_i \cdot A_j) + D_\alpha D_\beta (k_j \cdot A^i) W_j^\beta - D_\alpha W_i^\beta D_\beta (k_i \cdot A_j) \\
&\quad + D_\alpha D_\beta W_i^\sigma D_\sigma W_j^\beta - (D_\alpha W_i A_j) + (D_\alpha A^i W_j) + \frac{3}{2} D_\alpha F_{mn}^i F_j^{mn} \\
&= -k_m^j (\gamma^m W_i)_\alpha (k_i \cdot A_j) - (\gamma^m W_i)_\alpha A_m^j - A_\alpha^i (k_i \cdot A_j) + \frac{1}{4} k_m^j F_{st}^i (\gamma^{st} \gamma^m W_j)_\alpha \\
&\quad - \frac{1}{4} F_{st}^i k_m^i (\gamma^{st} \gamma^m W_j)_\alpha + 4k_j^t (\gamma_t W^i)_\alpha (k_i \cdot A_j) - \frac{1}{4} F_{mn}^i (\gamma^{mn} A_j)_\alpha + (D_\alpha A^i W_j) + 3k_m^i (\gamma_n W_i)_\alpha F_j^{mn} \\
&= -k_m^j (\gamma^m W_i)_\alpha (k_i \cdot A_j) - (\gamma^m W_i)_\alpha A_m^j - A_\alpha^i (k_i \cdot A_j) + 4k_j^t (\gamma_t W^i)_\alpha (k_i \cdot A_j) \\
&\quad - \frac{1}{4} F_{mn}^i (\gamma^{mn} A_j)_\alpha + (D_\alpha A^i W_j) - 3k_j^n (\gamma_n W_i)_\alpha (k^i \cdot A_j)
\end{aligned}$$

where, we can show that $-\frac{1}{4} F_{st}^i k_m^i (\gamma^{st} \gamma^m W_j)_\alpha = 0$. Consequently,

$$D_\alpha^i \tilde{U}^{ij} = -(\gamma^m W_i)_\alpha A_m^j - A_\alpha^i (k_i \cdot A_j) - \frac{1}{4} F_{mn}^i (\gamma^{mn} A_j)_\alpha + (D_\alpha A^i W_j)$$

With this we find that:

$$\begin{aligned}
A_\alpha^{ij} &= -A_m^i (\gamma^m W_j)_\alpha - (k_j \cdot A_i) A_\alpha^j + (\gamma^m W_i)_\alpha A_m^j - W_i^\beta D_\beta A_\alpha^j + A_\alpha^i (k_i \cdot A_j) + D_\beta A_\alpha^i W_j^\beta \\
&\quad - (\gamma^m W_i)_\alpha A_m^j - A_\alpha^i (k_i \cdot A_j) - \frac{1}{4} F_{mn}^i (\gamma^{mn} A_j)_\alpha + (D_\alpha A^i W_j) \\
&= -A_m^i (\gamma^m W_j)_\alpha - (k_j \cdot A_i) A_\alpha^j - W_i^\beta D_\beta A_\alpha^j + D_\beta A_\alpha^i W_j^\beta - \frac{1}{4} F_{mn}^i (\gamma^{mn} A_j)_\alpha + (D_\alpha A^i W_j) \\
&= -(k_j \cdot A_i) A_\alpha^j - W_i^\beta D_\beta A_\alpha^j - \frac{1}{4} F_{mn}^i (\gamma^{mn} A_j)_\alpha
\end{aligned}$$

(36) {?}

Next we calculate the various contributions in (30)

$$\begin{aligned}
(A_1 \cdot A_{23}) &= A_1^m \left(A_m^2 (k_2 \cdot A_3) - (k_3 \cdot A_2) A_m^3 - D_\alpha A_m^2 W_3^\alpha - W_2^\alpha D_\alpha A_m^3 + (W_2 \gamma_m W_3) + k_m^2 \tilde{U}^{23} \right) \\
&= (A^1 \cdot A^2) (k_2 \cdot A_3) - (k_3 \cdot A_2) (A^1 \cdot A^3) - A_1^m D_\alpha A_m^2 W_3^\alpha - A_1^m W_2^\alpha D_\alpha A_m^3 + A_1^m (W_2 \gamma_m W_3) \\
&\quad + (k_2 \cdot A_1) \left[- (k_3 \cdot A_2) (k_2 \cdot A_3) - (A_2 \cdot A_3) + D_\alpha (k_3 \cdot A^2) W_3^\alpha - W_2^\alpha D_\alpha (k_2 \cdot A_3) \right. \\
&\quad \left. + D_\beta W_2^\alpha D_\alpha W_3^\beta + \textcolor{blue}{k}_m^{23} (\textcolor{blue}{W}_2 \gamma^m \textcolor{blue}{W}_3) - (W_2 A_3) + (A^2 W_3) + \frac{3}{2} F_{mn}^2 F_3^{mn} \right] \\
&= (A^1 \cdot A^2) (k_2 \cdot A_3) - (k_3 \cdot A_2) (A^1 \cdot A^3) - \textcolor{red}{A}_1^m (\textcolor{red}{W}_2 \gamma_m \textcolor{red}{W}_3) - (k_2 \cdot A_1) (A_2 W_3) \\
&\quad - A_1^m (W_2 \gamma_m W_3) - (k_3 \cdot A_1) (W_2 A^3) + \textcolor{red}{A}_1^m (\textcolor{red}{W}_2 \gamma_m \textcolor{red}{W}_3) \\
&\quad + (k_2 \cdot A_1) \left[- (k_3 \cdot A_2) (k_2 \cdot A_3) - (A_2 \cdot A_3) + \textcolor{blue}{D}_\alpha (\textcolor{blue}{k}_3 \cdot A^2) W_3^\alpha - W_2^\alpha D_\alpha (k_2 \cdot A_3) \right. \\
&\quad \left. + \frac{1}{16} (\gamma^{mn})_\beta^\alpha (\gamma_{st})_\beta^\alpha F_{mn}^1 F_2^{st} - (W_2 A_3) + (A^2 W_3) + \frac{3}{2} F_{mn}^2 F_3^{mn} \right] \\
&= (A^1 \cdot A^2) (k_2 \cdot A_3) - (k_3 \cdot A_2) (A^1 \cdot A^3) - (\textcolor{red}{k}_2 \cdot \textcolor{red}{A}_1) (\textcolor{red}{A}_2 \textcolor{red}{W}_3) - A_1^m (W_2 \gamma_m W_3) \\
&\quad - (\textcolor{blue}{k}_3 \cdot \textcolor{blue}{A}_1) (\textcolor{blue}{W}_2 A^3) + (k_2 \cdot A_1) \left[- (k_3 \cdot A_2) (k_2 \cdot A_3) - (A_2 \cdot A_3) \right. \\
&\quad \left. - (\textcolor{blue}{W}_2 A_3) + (\textcolor{red}{A}^2 \textcolor{red}{W}_3) + (k_3 \cdot A_2) (k_2 \cdot A_3) \right]
\end{aligned}$$

where, we used

$$\frac{1}{16} (\gamma^{mn})_\beta^\alpha (\gamma_{st})_\beta^\alpha F_{mn}^1 F_2^{st} + \frac{3}{2} F_{mn}^2 F_3^{mn} = -2 F_{mn}^2 F_3^{mn} + \frac{3}{2} F_{mn}^2 F_3^{mn} = (k_3 \cdot A_2) (k_2 \cdot A_3) \quad (37) \{?\}$$

and that the terms in blue in the last equality vanish by $((k_3 + k_2) \cdot A_1) = -(k_1 \cdot A_1) = 0$

Hence, we find

$$(A_1 \cdot A_{23}) = (A^1 \cdot A^2) (k_2 \cdot A_3) - (k_3 \cdot A_2) (A^1 \cdot A^3) - A_1^m (W_2 \gamma_m W_3) - (k_2 \cdot A_1) (A_2 \cdot A_3) \quad (38) \{?\}$$

To calculate the next contrition $W_1^\alpha D_\alpha (k_1 \cdot A_{23})$, we need

$$\begin{aligned}
D_\alpha A_m^{ij} &= D_\alpha A_m^i (k_i \cdot A_j) + A_m^i (k_i \cdot D_\alpha A^j) - D_\alpha (D_\beta A_m^i) W_j^\beta + (D_\beta A_m^i) D_\alpha W_j^\beta \\
&\quad + \frac{1}{2} (D_\alpha W_i \gamma_m W_j) - \frac{1}{2} (W_i \gamma_m D_\alpha W_j) - (i \leftrightarrow j) \\
&= [(\gamma_m W^i)_\alpha + k_m^i A_\alpha^i] (k_i \cdot A_j) + A_m^i k_n^i [(\gamma^n W_j)_\alpha + k_j^n A_\alpha^j] - D_\alpha [(\gamma_m W_i)_\beta + k_m^i A_\beta^i] W_j^\beta \\
&\quad + \frac{1}{4} [(\gamma_m W^i)_\beta + k_m^i A_\beta^i] (\gamma^{st})_\alpha^\beta F_{st}^j + \frac{1}{8} (F_{pq}^i (\gamma^{pq} \gamma^m W_j)_\alpha) - \frac{1}{8} (W_i \gamma_m)_\beta (\gamma^{pq})_\alpha^\beta F_{pq}^j - (i \leftrightarrow j) \quad (39) \{?\}
\end{aligned}$$

Thus,

$$\begin{aligned}
W_1^\alpha D_\alpha (k_1 \cdot A_{23}) &= W_1^\alpha k_1^m D_\alpha A_m^{23} = [k_1^m (W_1 \gamma_m W_2) + (k_1 \cdot k_2) (W_1 A^2)] (k_2 \cdot A_3) \\
&+ (k_1 \cdot A_2) k_n^2 [(W_1 \gamma^n W_3) + k_3^n (W_1 A^3)] \\
&- \left[k_1^m W_1^\alpha (\gamma_m)_{\beta\sigma} D_\alpha W_2^\sigma + k_1^m k_m^2 W_1^\alpha D_\alpha A_\beta^2 \right] W_3^\beta \\
&+ \frac{1}{4} \left[k_1^m W_1^\alpha (\gamma_m W^2)_\beta + k_1^m k_m^2 W_1^\alpha A_\beta^2 \right] (\gamma^{st})_\alpha^\beta F_{st}^j + \frac{1}{8} (F_{pq}^2 k_1^m (W_1 \gamma^{pq} \gamma^m W_3)) \\
&- \frac{1}{8} k_1^m (W_1 \gamma^{pq} \gamma^m W_2) F_{pq}^3 - (i \leftrightarrow j) \\
&= -\frac{1}{4} k_1^m (W_1 \gamma^{st} \gamma^m W_3) F_{st}^2 + \frac{1}{4} k_1^m (W_1 \gamma^{st} \gamma^m W_2) F_{st}^3 + \frac{1}{8} (F_{pq}^2 k_1^m (W_1 \gamma^{pq} \gamma^m W_3)) \\
&- \frac{1}{8} k_1^m (W_1 \gamma^{pq} \gamma^m W_2) F_{pq}^3 - (i \leftrightarrow j) \\
&= -\frac{1}{8} k_1^m (W_1 \gamma^{st} \gamma^m W_3) F_{st}^2 + \frac{1}{8} k_1^m (W_1 \gamma^{st} \gamma^m W_2) F_{st}^3 - (i \leftrightarrow j) \tag{40} \{?\}
\end{aligned}$$

Note that using $\gamma^{pq}\gamma^m = \gamma^{pqm} + \gamma^p\eta^{mq} - \gamma^q\eta^{mp}$, we find that

$$\begin{aligned}
k_1^m (W_1 \gamma^{st} \gamma^m W_3) F_{st}^2 &= k_1^m (W_1 \gamma^{stm} W_3) F_{st}^2 + k_1^t (W_1 \gamma^s W_3) F_{st}^2 - k_1^s (W_1 \gamma^t W_3) F_{st}^2 \\
&= 2k_1^t (W_1 \gamma^s W_3) F_{st}^2 = 2k_1^t (W_1 \gamma^s W_3) k_s^2 A_t^2 - 2k_1^t (W_1 \gamma^s W_3) k_t^2 A_s^2 = 0
\end{aligned}$$

and similarly $2 \leftrightarrow 3$ implies

$$k_1^m (W_1 \gamma^{st} \gamma^m W_2) F_{st}^3 = 0 \tag{41} \{?\}$$

Consequently,

$$W_1^\alpha D_\alpha (k_1 \cdot A_{23}) = 0$$

Next let us calculate

$$\begin{aligned}
D_\alpha W_{ij}^\beta &= D_\alpha \left[- (k_j \cdot A_i) W_j^\beta + D_\sigma W_i^\beta W_j^\sigma - (i \leftrightarrow j) \right] \\
&= -k_j^m [(\gamma_m W_i)_\alpha + k_m^i A_\alpha^i] W_j^\beta - \frac{1}{4} (k_j \cdot A_i) (\gamma^{mn})_\alpha^\beta F_{mn}^j + \frac{1}{4} (\gamma^{mn})_\sigma^\beta D_\alpha F_{mn}^i W_j^\sigma \\
&\quad + \frac{1}{16} (\gamma^{mn})_\sigma^\beta F_{mn}^i (\gamma^{st})_\alpha^\sigma F_{st}^j \\
&= -k_j^m (\gamma_m W_i)_\alpha W_j^\beta - \frac{1}{4} (k_j \cdot A_i) (\gamma^{mn})_\alpha^\beta F_{mn}^j + \frac{1}{4} (\gamma^{mn})_\sigma^\beta (2k_{[m}^i (\gamma_{n]} W^i)_\alpha) W_j^\sigma \\
&\quad + \frac{1}{16} F_{mn}^i (\gamma^{st} \gamma^{mn})_\alpha^\beta F_{st}^j \tag{42} \{?\}
\end{aligned}$$

Thus,

$$\begin{aligned}
D_\beta W_1^\alpha D_\alpha W_{23}^\beta &= \frac{1}{4} (\gamma^{pq})_\beta^\alpha F_{pq}^1 D_\alpha W_{23}^\beta \\
&= \frac{1}{4} k_3^m F_{pq}^1 (W_2 \gamma_m \gamma^{pq} W_3) - \frac{1}{4} F_{pq}^1 \text{Tr} (\gamma^{pq} \gamma^{mn}) (k_3 \cdot A_2) F_{mn}^j \\
&\quad + \frac{1}{8} (\gamma^{mn} \gamma^{pq})_\sigma^\alpha F_{pq}^1 (k_m^2 (\gamma_n W^2)_\alpha) W_3^\sigma + \frac{1}{64} F_{pq}^1 F_{mn}^2 \text{Tr} (\gamma^{pq} \gamma^{st} \gamma^{mn}) F_{st}^3 \\
&= -\frac{1}{4} k_3^m F_{pq}^1 (W_3 \gamma^{pq} \gamma_m W_2) - \frac{1}{4} F_{pq}^1 (-32 \delta_{mn}^{pq}) (k_3 \cdot A_2) F_3^{mn} \\
&\quad + \frac{1}{8} F_{pq}^1 k_m^2 (W_2 \gamma_n \gamma^{pq} \gamma^{mn} W_3) + \frac{1}{64} F_{pq}^1 F_{mn}^2 \text{Tr} (\gamma^{pq} \gamma^{st} \gamma^{mn}) F_{st}^3 \\
&= -16 (k_3 \cdot A_1) (k_3 \cdot A_2) (k_1 \cdot A_3) + 4 (k_3 \cdot A_1) (k_1 \cdot A_2) (k_2 \cdot A_3) \\
&= -12 (k_3 \cdot A_1) (k_1 \cdot A_2) (k_2 \cdot A_3)
\end{aligned} \tag{43} \{?\}$$

$$\begin{aligned}
(W_1 A_{23}) &= W_1^\alpha \left(- (k_j \cdot A_i) A_\alpha^j - W_i^\beta D_\beta A_\alpha^j - \frac{1}{4} F_{mn}^i (\gamma^{mn} A_j)_\alpha \right) \\
&= - (k_3 \cdot A_2) (W_1 A^3) - W_1^\alpha W_2^\beta D_\beta A_\alpha^3 - \frac{1}{4} F_{mn}^2 (W_1 \gamma^{mn} A_3)
\end{aligned} \tag{44} \{?\}$$

$$\begin{aligned}
(A^1 W_{23}) &= A_1^\alpha W_{23}^\alpha = A_1^\alpha \left(- (k_3 \cdot A_2) W_3^\alpha + W_2^\alpha (k_2 \cdot A_3) + D_\beta W_2^\alpha W_3^\beta - W_2^\beta D_\beta W_3^\alpha \right) \\
&= - (k_3 \cdot A_2) (A_1 W_3) + (A_1 W_2) (k_2 \cdot A_3) - \frac{1}{4} (A_1 \gamma^{mn} W_3) F_{mn}^2 + \frac{1}{4} (A_1 \gamma^{mn} W_2) F_{mn}^3
\end{aligned} \tag{45} \{?\}$$

$$\begin{aligned}
F_1^{mn} F_{mn}^{23} &= F_1^{mn} \left[- (k_3 \cdot A_2) F_{mn}^3 - W_2^\alpha D_\alpha F_{mn}^3 + F_{mn}^2 (k_2 \cdot A_3) - D_\alpha F_{mn}^2 W_3^\alpha - 2 F_{mp}^2 F_3^p{}_{n} \right] \\
&= - (k_3 \cdot A_2) F_1^{mn} F_{mn}^3 - F_1^{mn} W_2^\alpha (2 k_{[m}^3 (\gamma_{n]} W_3)_\alpha) + F_1^{mn} F_{mn}^2 (k_2 \cdot A_3) \\
&\quad - F_1^{mn} (2 k_{[m}^2 (\gamma_{n]} W_2)_\alpha) W_3^\alpha - 2 F_1^{mn} F_{mp}^2 F_3^p{}_{n} \\
&= 2 (k_3 \cdot A_1) (k_1 \cdot A_2) (k_2 \cdot A_3) - 2 F_1^{mn} k_m^3 (W_2 \gamma_n W_3) + 2 (k_3 \cdot A_1) (k_1 \cdot A_2) (k_2 \cdot A_3) \\
&\quad - 2 F_1^{mn} k_m^2 (W_2 \gamma_n W_3) - 4 (k_3 \cdot A_1) (k_1 \cdot A_2) (k_2 \cdot A_3) \\
&= 0
\end{aligned} \tag{46} \{?\}$$

Hence, we find on adding all the contributions

$$\begin{aligned}
S_{1(23)}^{II} &= \frac{1}{z_{23}z_{13}^2} \left[- \left((A^1 \cdot A^2) (k_2 \cdot A_3) - (k_3 \cdot A_2) (A^1 \cdot A^3) - A_1^m (W_2 \gamma_m W_3) - (k_2 \cdot A_1) (A_2 \cdot A_3) \right) \right. \\
&\quad - 12 (k_3 \cdot A_1) (k_1 \cdot A_2) (k_2 \cdot A_3) \\
&\quad - \left(- (k_3 \cdot A_2) (W_1 A^3) - W_1^\alpha W_2^\beta D_\beta A_\alpha^3 - \frac{1}{4} F_{mn}^2 (W_1 \gamma^{mn} A_3) \right) \\
&\quad \left. - (k_3 \cdot A_2) (A_1 W_3) + (A_1 W_2) (k_2 \cdot A_3) - \frac{1}{4} (A_1 \gamma^{mn} W_3) F_{mn}^2 + \frac{1}{4} (A_1 \gamma^{mn} W_2) F_{mn}^3 \right] \\
&= \frac{1}{z_{23}z_{13}^2} \left[- (A^1 \cdot A^2) (k_2 \cdot A_3) + (k_3 \cdot A_2) (A^1 \cdot A^3) + A_1^m (W_2 \gamma_m W_3) + (k_2 \cdot A_1) (A_2 \cdot A_3) \right. \\
&\quad - 12 (k_3 \cdot A_1) (k_1 \cdot A_2) (k_2 \cdot A_3) + (k_3 \cdot A_2) (W_1 A^3) + W_1^\alpha W_2^\beta D_\beta A_\alpha^3 + \frac{1}{4} F_{mn}^2 (W_1 \gamma^{mn} A_3) \\
&\quad \left. - (k_3 \cdot A_2) (A_1 W_3) + (A_1 W_2) (k_2 \cdot A_3) - \frac{1}{4} (A_1 \gamma^{mn} W_3) F_{mn}^2 + \frac{1}{4} (A_1 \gamma^{mn} W_2) F_{mn}^3 \right]
\end{aligned} \tag{47} \{?\}$$

References